

LECTURE 9: MATH1061/7861
TOPICS: PROOF BY CONTRADICTION
DATE: THURSDAY MARCH 12, 2026

Learning objectives: Learn how to write a proof by contradiction.



Reductio ad absurdum, which Euclid loved so much, is one of a mathematician's finest weapons. It is a far finer gambit than any chess play: a chess player may offer the sacrifice of a pawn or even a piece, but a mathematician offers the game.

— G. H. Hardy —

AZ QUOTES

Suppose want to prove the statement
(if p then q .)

Suppose for the sake of contradiction, we
have p but not q . See if you can
reach a contradiction.

Q1: Use proof by contradiction to show:

$\forall n \in \mathbb{Z}$, if $3n^3 - 2$ is odd then n is odd.

Suppose for the sake of contradiction that the statement is false. This means

$\exists n \in \mathbb{Z}$ s.t. $3n^3 - 2$ is odd but n is even.

Now, n being even means $n = 2k$ for some $k \in \mathbb{Z}$.

This implies $3n^3 - 2 = 3(2k)^3 - 2$

$$= 3(8k^3) - 2$$

$$= 24k^3 - 2$$

$$= 2(12k^3 - 1)$$

This means $3n^3 - 2$ is even, which is a contradiction.

Q2: Use proof by contradiction to show:

$\forall m, n \in \mathbb{Z}$, if mn is even then either m is even or n is even.

Suppose, for the sake of contradiction, the statement is false. This means there exists $m, n \in \mathbb{Z}$ s.t. mn is even but m and n are both odd. This then implies $m = 2k+1$ and $n = 2l+1$ for some integers k, l . Now we have

$$\begin{aligned} mn &= (2k+1)(2l+1) \\ &= 4kl + 2k + 2l + 1 \\ &= 2(2kl + k + l) + 1 \end{aligned}$$

So mn is odd, which is a contradiction.

Euclid's Proof of Infinitude of Primes:

Suppose, for the ~~st~~ sake of contradiction, there exist finitely many primes. Then we can write them in a finite list $P_1, P_2, \dots, P_n$.

Consider the number $x = P_1 P_2 P_3 \dots P_n - 1$.

Let's call this number x . Note $x > P_i$ for all $i \in \{1, 2, \dots, n\}$.
If x is a prime then we have a new prime not in the list $P_1, P_2, \dots, P_n$. This is a contradiction.

Suppose x is not a prime. Let P be a prime factor of x . We claim $P \neq P_i$ for any i . This would be a contradiction.

To see the claim observe that P_i

is a factor of $x+1 = P_1 \dots P_n$.

If P_i was also a factor of x , then it would have to be a factor of $(x+1) - x = 1$.

This is a contradiction because 1 doesn't have any prime factors.

A cleaner exposition of proof of infinitude of primes.

Suppose, for the sake of contradiction,

$\exists$ finitely many primes $p_1, p_2, \dots, p_n$.

Consider the number $x = p_1 \times p_2 \times \dots \times p_n - 1$.

Let P be a prime factor of x .

We claim that $P \neq p_i$ for any i .

Indeed, $p_i \mid p_1 \times p_2 \times \dots \times p_n = x + 1$.

If $p_i \mid x$ then $p_i \mid [(x+1) - x] \Rightarrow p_i \mid 1$

This is a contradiction, which proves the claim.

So we see that P is ~~not~~ a prime number which is not in the list $p_1, \dots, p_n$.

This is a contradiction.

Thus, we must have infinitely many primes.